Singapore School Kinderland Secondary-2

Computer Science — Term 2 Comprehensive Study Material & Syllabus Guide

Cambridge IGCSE & O Level Computer Science (Syllabus 0478 / 2210)

Prescribed Textbook: Cambridge IGCSE Computer Science (2nd Edition)

Chapters Covered: Chapter 1.3 (Pg 32–44) & Chapter 2.2, 2.3 (Pg 54–74)

Academic Level: Secondary 2 (Grade 8)

Examination Scope: Class Test 1, Class Test 2 & Term 2 Final Exams

MODULE 1: CHAPTER 1.3 DATA STORAGE & FILE SIZE CALCULATIONS (TB PG 32–40)

1.3.1 Measurement of Data Storage & Prefixes (TB Pg 32–34)

Fundamental Storage Units:

- **Bit (b):** Smallest unit of data representation, binary 0 or 1.
- **Nibble:** A group of 4 bits (e.g. 1010).
- **Byte (B):** A group of 8 bits. Capable of representing one ASCII character.

Metric (SI) Prefixes (Powers of 10): Used by storage manufacturers: 1 Kilobyte (KB) = 1,000 Bytes (10^3).
IEC Binary Prefixes (Powers of 2): Standard computer internal architecture: 1 Kibibyte (KiB) = 1,024 Bytes (2^{10}).

Unit Name	IEC Binary Symbol	Exact Power of 2 Value	Traditional SI Equivalent
Kibibyte	KiB	1,024 Bytes (2^{10})	1 KB = 1,000 Bytes (10^3)
Mebibyte	MiB	1,024 KiB ($2^{20} = 1,048,576$ B)	1 MB = 1,000 KB (10^6)
Gibibyte	GiB	1,024 MiB ($2^{30} = 1,073,741,824$ B)	1 GB = 1,000 MB (10^9)
Tebibyte	TiB	1,024 GiB (2^{40})	1 TB = 1,000 GB (10^{12})

1.3.2 Image File Size Calculations (TB Pg 35–37)

Bitmapped images are made up of individual picture elements called **pixels**. The size of an uncompressed image depends on two factors:

- **Image Resolution:** Number of pixels in width × height (e.g., 1024 × 768 pixels).